



NBA 5090 - Advanced Financial Statement Analysis

Financial Statements Forecasting and Valuation Analysis

Jasmine Lin (jl4772)

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Part 1: Brief Introduction

Deluxe Corporation (DLX) is a U.S.-based business services company operating four core segments: Merchant Services, B2B Payments, Data Solutions, and Print. Experiencing a continuous transformation from traditional check printing, it is evolving into a payment and data technology firm focused on digital payments and data solutions. Revenue from payment and data-related businesses already accounts for a significant proportion of total revenue, with financial services and print product revenue nearly equal. As stated in 10-K, the overall strategy is Leverage the cash flows, customer relationships, and brand equity from our Print business to fuel profitable organic growth in our other businesses. The company is projected to increase investment in financial services in the future.

Part 2: Forecasting and Projected Financial Statements

1. Income Statement

The complete income statement is presented below.

Income Statement								
For the Fiscal Period Ending	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
Revenues								
Product Revenues	1,286,197	1,257,600	1,205,374	1,193,320	1,175,420	1,151,912	1,128,874	1,106,296
Service Revenues	951,813	934,660	916,387	934,715	953,409	991,545	1,031,207	1,072,456
Total Revenues	2,238,010	2,192,260	2,121,761	2,128,035	2,128,829	2,143,457	2,160,081	2,178,752
Expenses								
Cost of Services	(561,879)	(543,548)	(540,170)	(541,250)	(542,874)	(545,046)	(547,771)	(551,057)
Cost of Products	(470,237)	(486,029)	(455,141)	(454,231)	(453,095)	(451,736)	(450,155)	(448,354)
Selling, General and Administrative Expenses	(993,250)	(956,068)	(909,168)	(901,895)	(894,679)	(890,206)	(885,755)	(881,326)
Restructuring and Integration Expenses	(62,529)	(78,245)	(48,570)	(20,000)	(8,000)	(5,000)	(3,000)	(2,000)
Interest Expense	(94,454)	(125,643)	(123,281)	(124,881)	(124,881)	(124,881)	(14,530)	(131,330)
Other Income	9,386	4,651	7,602	7,213	7,213	7,213	7,213	7,213
Asset Impairment Charges	-	-	(7,743)	0	0	0	0	0
Net Gain/loss on Sale of Facility	19,331	32,421	31,207	27,653	27,653	27,653	27,653	27,653
Earnings before Taxes	84,378	39,799	76,497	120,644	140,166	161,455	293,737	199,550
Taxes and Other Expenses								
Provision for Income Tax	(18,848)	(13,572)	(23,552)	(35,078)	(40,754)	(46,944)	(85,406)	(58,021)
Minority Interest (After Tax)	(135)	(107)	(143)	(252)	(293)	(337)	(614)	(417)
Net Income (Loss)	65,395	26,120	52,802	85,314	99,119	114,173	207,717	141,113

(1) Discussion of Revenue Forecasting

➤ Product revenue

- Deluxe's traditional Print business is a slowly declining segment. As stated in the company's strategy at 10-K, it aims to gradually transform into a fintech company and focus more on providing services to clients. Thus, I forecast Deluxe will gradually reduce the proportion of its Print business and increase support for services.
- However, nearly half of its business is related to checks, a stable segment. Although the printing industry is a mature market with a long-term downward trend (due to declining check usage), Deluxe remains one of the largest check printers in the U.S., boasting high market share, strong bargaining power, and significant economies of scale. While consumer check usage has trended downward amid digital payment adoption, corporate and government clients still rely on checks for compliance needs (e.g., audit trails, formal payment documentation), creating sustained demand. Therefore, these advantages mean the Print business's decline rate is projected to remain mild (estimated at 1-2% annually) rather than steep.

➤ *Service revenue*

In recent years, management has been driving a major transformation from a printing company to a payment and data company.

- The 2021 acquisition of First American Payment Systems marked Deluxe a significant stride into the payments sector. FAPS brought a scalable merchant payment processing platform, a client base of over thousands SMBs, and expertise in in-store and online payment acceptance
- Deluxe launched the "North Star" strategy in 2023, focusing resources on high-growth segments such as payments and data, while improving cash flow and reducing leverage through layoffs, backend integration, global outsourcing, etc.
- Since 2024, the company has reported under four new business segments, aligning more closely with the post-transformation business structure.

Merchant Services is a growth-oriented business. With the company shifting its focus to services, I anticipate slight growth in the next two years and moderate long-term growth.

Overall, due to economic gravity, the growth in service revenue will be offset by the decline in traditional Print business. As a stable enterprise, Deluxe's overall business growth will therefore proceed steadily in line with U.S. economic growth.

Income Statement								
For the Fiscal Period Ending								
Units	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
Revenues								
Product Revenues	1,286,197	1,257,600	1,205,374	1,193,320	1,175,420	1,151,912	1,128,874	1,106,296
				-1%	-1.50%	-2%	-2%	-2%
Service Revenues	951,813	934,660	916,387	934,715	953,409	991,545	1,031,207	1,072,456
				2%	2%	4%	4%	4%
Total Revenues	2,238,010	2,192,260	2,121,761	2,128,035	2,128,829	2,143,457	2,160,081	2,178,752

(2) Discussion of Expense Forecasting

➤ *COGS*

• *Cost of service*

Under the North Star initiative launched in 2023, Deluxe implements organizational flattening, back-office consolidation, increased automation and global staffing, and continues to close/consolidate certain facilities. Cost savings of about \$10 million in COGS and \$25 million in SG&A were achieved in 2024 via layoffs and other measures, with additional savings expected in 2025. As Merchant Services expands more embedded payment channels and Data Solutions penetrates additional industry verticals, Deluxe may invest more on such segment for a long term.

• *Cost of product*

The company's cost base broadly moves in line with revenue, but we expect a gradual improvement in gross margin over the forecast period. Management has already begun to execute on a series of cost-efficiency initiatives, including streamlining headcount, optimizing production processes, consolidating vendors, and tightening discretionary spending. These actions should allow total operating costs to grow more slowly than revenue, supporting moderate margin expansion without relying on aggressive top-line assumptions.

➤ SG&A

Under the North Star, cost savings of about \$10 million in COGS and \$25 million in SG&A were achieved in 2024 via layoffs and other measures, with additional savings expected in 2025.

➤ Restructuring and integration expenses

At the end of 2024, approximately \$95 million in restructuring & integration costs have been incurred, with a further \$15 million projected for 2025, primarily covering consulting fees, layoffs, and facility restructuring. This is not a one-time cost, but rather a recurring restructuring and integration expense for Deluxe amid its long-term transformation. It should be projected instead of to be excluded.

➤ Interest expense

Since additional borrowings are assumed for 2025 and 2029 (See B/S forecasting below), and the company will repay substantial debt in 2028 (stated in the 10-K), interest expense will decrease significantly in 2028 while remaining stable in other years.

Income Statement								
For the Fiscal Period Ending	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
Expenses								
Cost of Services	(561,879)	(543,548)	(540,170)	(541,250)	(542,874)	(545,046)	(547,771)	(551,057)
				0.2%	0.30%	0.40%	0.5%	0.6%
Cost of Products	(470,237)	(486,029)	(455,141)	(454,231)	(453,095)	(451,736)	(450,155)	(448,354)
				-0.20%	-0.25%	-0.30%	-0.35%	-0.40%
Selling, General and Administrative Expenses	(993,250)	(956,068)	(909,168)	(901,895)	(894,679)	(890,206)	(885,755)	(881,326)
				-0.80%	-0.80%	-0.50%	-0.50%	-0.50%
Restructuring and Integration Expenses	(62,529)	(78,245)	(48,570)	(20,000)	(8,000)	(5,000)	(3,000)	(2,000)
Interest Expense	(94,454)	(125,643)	(123,281)	(124,881)	(124,881)	(124,881)	(14,530)	(131,330)
Other Income	9,386	4,651	7,602	7,213	7,213	7,213	7,213	7,213
Asset Impairment Charges	-	-	(7,743)	0	0	0	0	0
Net Gain/loss on Sale of Facility	19,331	32,421	31,207	27,653	27,653	27,653	27,653	27,653
Earnings before Taxes	84,378	39,799	76,497	120,644	140,166	161,455	293,737	199,550

(3) Discussion of Other Forecasting

I assume the company's effective tax rate is the simple average of its effective tax rates from 2022 to 2024. Given the assumed stability of the tax system environment in the near term, this approach ensures the model's overall precision, stabilizes the forecast, smooths out random fluctuations to a certain extent, better approximates the company's usual effective tax rate, and improves the forecasting model's accuracy.

In addition, I assume Minority interest is calculated as $\text{Minority Interest (After-Tax)} = \text{After-Tax Net Earnings} \times \text{Average Minority Interest Portion}$.

Income Statement								
For the Fiscal Period Ending	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
Taxes and Other Expenses								
Provision for Income Tax	(18,848)	(13,572)	(23,552)	(35,078)	(40,754)	(46,944)	(85,406)	(58,021)
	-22.34%	-34.10%	-30.79%	-29.08%	-29.08%	-29.08%	-29.08%	-29.08%
Minority Interest (After Tax)	(135)	(107)	(143)	(252)	(293)	(337)	(614)	(417)
	(0.002060)	(0.004080)	(0.002701)	(0.002947)	(0.002947)	(0.002947)	(0.002947)	(0.002947)
Net Income (Loss)	65,395	26,120	52,802	85,314	99,119	114,173	207,717	141,113

2. Balance Sheet

The complete balance sheet is presented below.

Balance Sheet								
Balance Sheet as of:	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
Current Assets								
Cash and Cash Equivalents	40,435	71,962	34,399	202,947	147,921	8,787	18,630	228,038
Trade Accounts Receivable,	206,617	191,005	174,076	185,488	185,557	186,832	188,281	189,909
Net of Allowance for Un Collectible Accounts								
Revenue in Excess of Billing	38,761	26,107	26,741	29,673	29,684	29,888	30,120	30,380
Inventories & Supplies	52,267	42,088	36,393	42,502	42,523	42,557	42,606	42,670
Prepaid Expenses	36,642	30,116	32,751	32,406	32,421	32,448	32,485	32,534
Other Current Assets	27,024	16,576	35,403	25,765	25,774	25,951	26,153	26,379
Funds Held for Customers	302,291	-	-	-	-	-	-	-
Settlement Processing Assets	-	383,134	271,876	322,295	297,598	312,137	308,262	314,102
Total Current Assets	704,037	760,988	611,639	841,074	761,479	638,601	646,537	864,010
Non Current Assets								
Operating Lease Assets	47,132	58,961	49,382	50,526	50,545	50,892	51,287	51,730
Property, Plant & Equipment,	124,894	116,539	111,587	191,486	261,885	322,785	374,184	416,083
Net of Accumulated Depreciation								
Long-term Investments	47,783	61,924	61,025	60,000	60,000	60,000	60,000	60,000
Deferred Income Taxes	1,956	8,694	6,969	5,763	5,765	5,805	5,850	5,900
Goodwill	1,431,385	1,430,590	1,422,737	1,412,737	1,402,737	1,392,737	1,382,737	1,372,737
Intangibles, Net	458,979	391,744	331,053	305,727	308,750	340,103	391,740	456,019
Other Non-current Assets	260,354	251,182	236,644	242,909	243,000	244,670	246,567	248,698
Total Assets	3,076,520	3,080,622	2,831,036	3,110,222	3,094,161	3,055,592	3,158,901	3,475,178
Current Liabilities								
Accounts Payable	157,055	154,863	164,878	155,374	155,450	155,577	155,755	155,987
Accrued Liabilities	218,404	191,427	149,593	181,175	181,243	182,488	183,904	185,493
Current Portion of Long-term Debt	71,748	86,153	37,130	37,500	116,400	50,000	1,280,500	40,000
Funds Held for Customers	305,138	-	-	-	-	-	-	-
Settlement Processing Obligations	-	386,622	273,915	322,295	297,598	312,137	308,262	314,102
Total Current Liabilities	752,345	819,065	625,516	696,344	750,691	700,202	1,928,422	695,582
Non Current Liabilities								
Long-term Debt	1,572,528	1,506,698	1,466,021	1,628,521	1,512,121	1,462,121	181,621	1,641,621
Operating Lease Liabilities	48,925	58,840	48,982	50,526	50,545	50,892	51,287	51,730
Deferred Income Taxes	45,510	22,649	2,104	22,456	22,465	22,619	22,795	22,992
Minority Interest	415	522	166	368	368	368	368	368
Other Long-term Liabilities	52,988	68,754	67,495	61,606	61,629	62,053	62,534	63,074
Shareholders' Equity								
Common Stock - Par Value	43,204	43,743	44,315	44,315	44,315	44,315	44,315	44,315
Additional Paid in Capital	79,234	99,141	117,122	117,122	117,122	117,122	117,122	117,122
Accumulated Deficit/retained Earnings	518,635	491,238	489,231	521,367	567,308	628,303	782,842	870,776
Accumulated Other Comprehensive Loss, Net of Tax	(37,264)	(30,028)	(29,916)	(32,403)	(32,403)	(32,403)	(32,403)	(32,403)
Total Shareholders Equity	603,809	604,094	620,752	650,401	696,342	757,337	911,876	999,811
Total Liabilities & Shareholders Equity	3,076,520	3,080,622	2,831,036	3,110,222	3,094,161	3,055,592	3,158,901	3,475,178
<i>Balance check</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>

(1) Discussion of Net Operating Assets Forecasting

➤ *Current assets*

- *Trade account receivables, revenue in excess billing, inventories and supplies, prepaid expenses, other current assets* are assumed to equal the average portion of sales from 2022–2024 multiplied by the current year's revenue. These accounts are highly correlated with revenue.
- *Funds held for customers* were replaced after the emergence of settlement processing assets, with subsequent balances being zero, matching the corresponding funds held for customers (liability).

Funds held for customers and settlement processing assets are both funds temporarily held on behalf of customers or in transit for clearing in the company's payment/settlement business.

- *Settlement processing assets* are calculated as the weighted moving average of the previous two years, matching the corresponding settlement processing obligation. Using weighted moving average ensures the alignment with sales and remains some extent to fluctuations.

➤ *Current liabilities*

- *Accounts payables, accrued liabilities* are assumed to equal the average portion of sales from 2022–2024 multiplied by the current year's COGS (sum of cost of services and cost of product). These accounts are highly correlated with COGS.
- *Current portion of long-term debt*

Through the balance sheet and Note 13 of Deluxe's 2024 10-K, long-term debt maturities as of December 31, 2024 are as follows: 2025: 37,500/ 2026: 37,500/ 2027: 116,417/ 2028: 50,000/ 2029: 1,280,500. A securitization facility will mature in 2027. The 2029 repayment obligation includes the remaining term loan, senior unsecured notes, senior secured notes, and revolver.

Balance Sheet								
Balance Sheet as of:	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
Current Assets								
Cash and Cash Equivalents	40,435	71,962	34,399	202,947	147,921	8,787	18,630	228,038
Trade Accounts Receivable,	206,617	191,005	174,076	185,488	185,557	186,832	188,281	189,909
Net of Allowance for Un Collectible Accounts								
Revenue in Excess of Billing	38,761	26,107	26,741	29,673	29,684	29,888	30,120	30,380
Inventories & Supplies	52,267	42,088	36,393	42,502	42,523	42,557	42,606	42,670
Prepaid Expenses	36,642	30,116	32,751	32,406	32,421	32,448	32,485	32,534
Other Current Assets	27,024	16,576	35,403	25,765	25,774	25,951	26,153	26,379
Funds Held for Customers	302,291	-	-	-	-	-	-	-
Settlement Processing Assets	-	383,134	271,876	322,295	297,598	312,137	308,262	314,102
Total Current Assets	704,037	760,988	611,639	841,074	761,479	638,601	646,537	864,010
Current Liabilities								
Accounts Payable	157,055	154,863	164,878	155,374	155,450	155,577	155,755	155,987
Accrued Liabilities	218,404	191,427	149,593	181,175	181,243	182,488	183,904	185,493
Current Portion of Long-term Debt	71,748	86,153	37,130	37,500	116,400	50,000	1,280,500	40,000
Funds Held for Customers	305,138	-	-	-	-	-	-	-
Settlement Processing Obligations	-	386,622	273,915	322,295	297,598	312,137	308,262	314,102
Total Current Liabilities	752,345	819,065	625,516	696,344	750,691	700,202	1,928,422	695,582

(2) Discussion of PP&E

In the "Purchases of capital assets" line item in Deluxe's 2024 10-K cash flow statement (2022: 104,598 / 2023: 100,747 / 2024: 94,389), the three-year average CAPEX is approximately 100 million. Salvage is assumed to be 5%. In the "Depreciation" line item in Deluxe's 2024 10-K cash flow statement (2022: 22,191 / 2023: 24,082 / 2024: 26,218), to remain at the depreciation level, the useful life of PPE should be assumed to be 10 years.

CAPEX Assumptions			
in thousands of USD	Cost	Salvage	Life
2025 Beginning PPE	111,587	5%	10
2025 CAPEX	100,000	5%	10
2026 CAPEX	100,000	5%	10
2027 CAPEX	100,000	5%	10
2028 CAPEX	100,000	5%	10
2029 CAPEX	100,000	5%	10

Depreciation Schedule					
	2025E	2026E	2027E	2028E	2029E
2025	10,601	10,601	10,601	10,601	10,601
2025	9,500	9,500	9,500	9,500	9,500
2026	-	9,500	9,500	9,500	9,500
2027	-	-	9,500	9,500	9,500
2028	-	-	-	9,500	9,500
2029	-	-	-	-	9,500
Total depreciation	20,101	29,601	39,101	48,601	58,101

The complete net PP&E schedule is presented below.

Net PPE Schedule					
	2025E	2026E	2027E	2028E	2029E
Beginning balance	111,587	191,486	261,885	322,785	374,184
Add: CAPEX	100,000	100,000	100,000	100,000	100,000
Less: Depreciation	-20,101	-29,601	-39,101	-48,601	-58,101
Ending balance	191,486	261,885	322,785	374,184	416,083

(3) Discussion of Intangible Assets

- In the note of intangible asset in Deluxe's 2024 10-K balance sheet (internal use software: 2022: 79,394 / 2023: 81,349 / 2024: 74,778), the three-year average CAPEX of intangible asset is approximately 84.6 million. I assume that the CAPEX is projected to be \$80 million per year as the projected amortization will be declining in the near future.
- Projected amortization data each year are given in the 10-K (2025: 105,326 / 2026: 76,977 / 2027: 48,647 / 2028: 28,363 / 2029: 15,721)
- According to historical data that goodwill is amortized at a similar level each year, I assume goodwill amortization is projected to be about \$20 million per year.

The complete net intangible asset schedule is presented below.

Net Intangible asset schedule					
	2025E	2026E	2027E	2028E	2029E
Beginning balance	331,053	305,727	308,750	340,103	391,740
Add: CAPEX	80,000	80,000	80,000	80,000	80,000
Less: Amortization	-105,326	-76,977	-48,647	-28,363	-15,721
Ending balance	305,727	308,750	340,103	391,740	456,019

(4) Discussion of Shareholder's Equity

➤ *Dividend*

Over the past three years, Deluxe's cash dividends have remained stable at \$0.30 per share per quarter, with four distributions annually, that is \$1.20 per share per year. Therefore, I assume this situation will remain the same in the near future.

Dividend					
	2025E	2026E	2027E	2028E	2029E
Share outstanding	44,315	44,315	44,315	44,315	44,315
Dividends per share	1.2	1.2	1.2	1.2	1.2
Dividends	53,178	53,178	53,178	53,178	53,178

➤ *Retained earnings*

The retained earnings calculation is presented as below.

Retained earnings calculation					
	2025E	2026E	2027E	2028E	2029E
Beginning balance	489,231	521,367	567,308	628,303	782,842
Add: Net income	85,314	99,119	114,173	207,717	141,113
Less: Dividend	-53,178	-53,178	-53,178	-53,178	-53,178
Ending balance	521,367	567,308	628,303	782,842	870,776

➤ *Shareholder's equity*

Balance Sheet								
Balance Sheet as of:	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
Shareholders' Equity								
Common Stock - Par Value	43,204	43,743	44,315	44,315	44,315	44,315	44,315	44,315
Additional Paid in Capital	79,234	99,141	117,122	117,122	117,122	117,122	117,122	117,122
Accumulated Deficit/retained Earnings	518,635	491,238	489,231	521,367	567,308	628,303	782,842	870,776
Accumulated Other Comprehensive Loss, Net of Tax	(37,264)	(30,028)	(29,916)	(32,403)	(32,403)	(32,403)	(32,403)	(32,403)
Total Shareholders Equity	603,809	604,094	620,752	650,401	696,342	757,337	911,876	999,811

I assume that common stock, additional paid in capital, accumulated other comprehensive loss will remain stable in the near future.

(5) Discussion of Financial Accounts

➤ *Long-term debt increase in 2025E, 2029E, increasing current portion of long-term debt and int. exp.*

- It is assumed that long-term borrowings will increase by \$20 million in 2025 to avoid insolvency due to the substantial debt maturity in 2029.
- A securitization facility will mature in 2027, and the 2029 repayment obligation includes the remaining term loan, senior unsecured notes, senior secured notes, and revolver. Thus, long-term debt will drop significantly in these two years.
- However, to maintain a stable debt level and support business transformation, it is assumed that the company will take on an additional \$1.5 billion in long-term borrowings in 2029. Therefore, the current portion of long-term debt and interest expense in I/S will increase accordingly.

➤ *Operating lease liabilities, deferred income taxes, other long-term liabilities* are assumed to equal the average portion of sales from 2022–2024 multiplied by the current year's revenue.

Balance Sheet								
Balance Sheet as of:	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
Current Liabilities								
Accounts Payable	157,055	154,863	164,878	155,374	155,450	155,577	155,755	155,987
	15.22%	15.04%	16.57%	15.61%	15.61%	15.61%	15.61%	15.61%
Accrued Liabilities	218,404	191,427	149,593	181,175	181,243	182,488	183,904	185,493
	9.76%	8.73%	7.05%	8.51%	8.51%	8.51%	8.51%	8.51%
Current Portion of Long-term Debt	71,748	86,153	37,130	37,500	116,400	50,000	1,280,500	40,000
Funds Held for Customers	305,138	-	-	-	-	-	-	-
Settlement Processing Obligations	-	386,622	273,915	322,295	297,598	312,137	308,262	314,102
Total Current Liabilities	752,345	819,065	625,516	696,344	750,691	700,202	1,928,422	695,582
Non Current Liabilities								
Long-term Debt	1,572,528	1,506,698	1,466,021	1,628,521	1,512,121	1,462,121	181,621	1,641,621
Operating Lease Liabilities	48,925	58,840	48,982	50,526	50,545	50,892	51,287	51,730
Deferred Income Taxes	45,510	22,649	2,104	22,456	22,465	22,619	22,795	22,992
	2.03%	1.03%	0.10%	1.06%	1.06%	1.06%	1.06%	1.06%
Minority Interest	415	522	166	368	368	368	368	368
Other Long-term Liabilities	52,988	68,754	67,495	61,606	61,629	62,053	62,534	63,074
	2.37%	3.14%	3.18%	2.89%	2.89%	2.89%	2.89%	2.89%

(6) Identification of Plug Account

To ensure Total Assets equals Total Liabilities and Shareholders' Equity, I used cash and cash equivalents as a plug account. As presented in the cash flow statement projections below, I also ensured that the cash figures on the balance sheet are articulated with those from the cash flow. The two financial statements will be consistent with each other.

3. Cash Flow Statement

After classifying balance sheet items into operating, investing, and financing activities, I prepared the cash flow statement based on their annual changes. Operating activities start with net income to common, adjusted for the non-cash item of depreciation and amortization (see fixed asset schedule and intangible asset schedule above). Investing activities incorporate CAPEX assumptions from the fixed asset and intangible asset schedules. Financing activities include cash dividend paid to shareholders.

After calculating the cash change, the year-end cash balance is equal to the plug account on the balance sheet.

The Cash Flow Statement is presented as below.

Cash Flow Statement					
CFS as of:	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands
Operating activities					
Net income	85,314	99,119	114,173	207,717	141,113
Depreciation	20,101	29,601	39,101	48,601	58,101
Amortization	105,326	76,977	48,647	28,363	15,721
Goodwill amortization	10,000	10,000	10,000	10,000	10,000
-ΔTrade Accounts Receivables	(11,412)	(69)	(1,275)	(1,449)	(1,627)
-ΔRevenue in Excess of Billing	(2,932)	(11)	(204)	(232)	(260)
-ΔInventories & Supplies	(6,109)	(21)	(35)	(49)	(63)
-ΔPrepaid Expenses	345	(16)	(26)	(37)	(48)
-ΔOther Current Assets	9,638	(10)	(177)	(201)	(226)
-ΔOperating Lease Assets	(1,144)	(19)	(347)	(395)	(443)
-ΔDeferred Income Taxes (asset)	1,206	(2)	(40)	(45)	(51)
ΔAccounts Payables	(9,504)	76	127	179	232
ΔAccrued Liabilities	31,582	68	1,245	1,415	1,590
ΔOperating Lease Liabilities	1,544	19	347	395	443
ΔDeferred Income Taxes (liabilities)	20,352	8	154	175	197
ΔOther Long-term Liabilities	(5,889)	23	423	481	541
Cash flow from operating activities	248,420	215,743	212,114	294,918	225,217
Investing activities					
-ΔAdditions to Property Plant and Equipment	(100,000)	(100,000)	(100,000)	(100,000)	(100,000)
-ΔLong-term Investments	1,025	0	0	0	0
-ΔAdditions to Intangible assets	(80,000)	(80,000)	(80,000)	(80,000)	(80,000)
-ΔOther Non-current Assets	(6,265)	(91)	(1,670)	(1,898)	(2,131)
Cash flow from investing activities	(185,240)	(180,091)	(181,670)	(181,898)	(182,131)
Financing activities					
ΔCurrent Portion of Long-term Debt	370	78,900	(66,400)	1,230,500	(1,240,500)
ΔLong-term Debt	162,500	(116,400)	(50,000)	(1,280,500)	1,460,000
ΔMinority Interest	202	0	0	0	0
ΔAccumulated Other Comprehensive Loss, Net of Tax	(2,487)	0	0	0	0
ΔNet Change in Settlement Processing Obligations	(2,039)	0	0	0	0
-ΔCash Dividends Paid to Shareholders	(53,178)	(53,178)	(53,178)	(53,178)	(53,178)
Cash flow from financing activities	105,368	(90,678)	(169,578)	(103,178)	166,322
Cash Change Calculation					
	2025E	2026E	2027E	2028E	2029E
Cash and cash equivalents at the beginning of the year	34,399	202,946	147,921	8,787	18,629
Net change in cash	168,547	(55,026)	(139,133)	9,842	209,408
Cash and cash equivalents at the end of the year	202,946	147,921	8,787	18,629	228,037
<i>Balance check</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>

4. Validity Checks

The Validity Checks verifies the rationality and internal consistency of model predictions through longitudinal comparison of key financial ratios from 2022 to 2024 and projected from 2025E to 2029E. Completed both historical and projected data shows all ratios fluctuate within a reasonable range.

Validity Check	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E
ROE	11.10%	1.42%	2.85%	4.90%	5.21%	5.93%	10.47%	6.79%
ROA	2.13%	0.42%	0.89%	1.44%	1.60%	1.86%	3.34%	2.13%
Asset turnover	72.77%	35.61%	35.89%	35.82%	34.31%	34.85%	34.76%	32.84%
Net income margin	2.92%	1.19%	2.49%	4.01%	4.66%	5.33%	9.62%	6.48%
EBITDA margin	15.70%	15.29%	17.22%	17.43%	17.46%	17.45%	17.83%	18.57%
EBIT margin	7.99%	7.55%	9.42%	11.54%	12.45%	13.36%	14.27%	15.19%

Part 3: Valuation Analysis

1. Terminal Value Modeling

WACC Calculation	
Equity risk premium	3.79%
5 year Equity beta (monthly)	1.39
Risk free rate (U.S. 10-Year Treasury Constant Maturity Rate)	4.10%
Cost of equity	9.37%
Cost of debt	7.70%
Effective tax rate	29.08%
After-tax cost of debt	5.46%
Stock price (12/01/2025)	20.09
Market Capitalization	904,200
Debt	1,503,151
Equity%	37.56%
Debt%	62.44%
WACC	6.93%
Terminal growth rate	3%

The cost of capital is calculated using the Weighted Average Cost of Capital (WACC) formula, which involves estimating the cost of equity, cost of debt, and their respective proportions in the total capital structure.

- The cost of equity is computed via the Capital Asset Pricing Model (CAPM). Key assumptions include: a market risk premium of 3.79% (based on Damodaran implied ERP (Dec 1, 2025, trailing 12-month with adjusted payout)), a beta coefficient of 1.39 (5-year beta from Capital IQ), and a risk-free rate of 4.10% (yield on U.S. 10-year Treasury bonds). Under these assumptions, the cost of equity is 9.37%.
- As of the end of 2024, the company's pre-tax weighted average cost of debt was 7.70%, with an assumed effective tax rate of 29.08% (weighted average). Thus, the after-tax cost of debt is 5.46%.
- The proportions of debt and equity in total capital are based on their market values. The equity market capitalization (obtained from Yahoo Finance) is \$942,000,000, and the total debt on the balance sheet is \$1,503,151,000. Accordingly, equity and debt account for 37.56% and 62.44% of total capital, respectively.

- Substituting all the above values into the WACC formula, the company's weighted average cost of capital is 6.93%.
- Deluxe's traditional Print business is mature and even slightly declining, while Merchant Services is growing but small in scale. As a mature company with slow transformation, its long-term overall growth has been relatively moderate. Therefore, my forecasting sets the growth rate close to the U.S. GDP growth rate and inflation rate, aligning its growth with the overall U.S. economy.

2. DCF Valuation

DCF Valuation Model						
	2024A	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
EBIT	199,778	245,525	265,047	286,336	308,266	330,880
Effective tax rate		29.08%	29.08%	29.08%	29.08%	29.08%
NOPAT		174,137	187,983	203,082	218,636	234,674
Add: Depreciation & Amortization		125,427	106,578	87,748	76,964	73,822
Less: CAPEX		(180,000)	(180,000)	(180,000)	(180,000)	(180,000)
Net working capital	(11,146)	(20,716)	(20,733)	(20,388)	(20,014)	(19,610)
Less: ΔNWC		9,570	17	(345)	(374)	(404)
FCF (firm)		129,134	114,578	110,484	115,225	128,092
PV factor	1.00	0.94	0.87	0.82	0.76	0.72
PV of FCF (firm)		120,767	100,210	90,369	88,140	91,633
PV of near-term FCF	491,119					
PV of terminal value	2,402,427					
Total firm value	2,893,546					
Less:						
LT debt	(1,466,021)					
ST debt	(37,130)					
Minority interest	(143)					
Add:						
Excess cash	(37,563)					
Value of common equity	1,352,689					
Share outstanding	44,315					
Forecasted per share price	\$ 30.52					
Current stock price (12/01/2025)	\$ 20.09					
Over/under valuation \$	-10.43					
Over/under valuation %	-34.18%					
RECOMMENDATION	BUY					

- This DCF valuation model is a tool for estimating company value via discounting future cash flows. To calculate Free Cash Flow: start from EBIT, deduct taxes to get NOPAT, add back depreciation and amortization, then subtract CAPEX and changes in net working capital to derive annual FCF.
- To calculate total firm value via discounted cash flow: use WACC as the discount factor to convert annual FCF into present value, sum the PV of near-term FCF, then add the PV of terminal value to get the total firm value.
- To calculate equity value and target share price: subtract long-term debt, short-term debt, and minority interest from total firm value, then add excess cash (assuming 1989 cash as the baseline for "operating cash requirement"), resulting in the value of common equity and then dividing this by shares outstanding to get the target share price of \$30.52.
- Valuation conclusion: Compared to the current stock price of \$20.09, the model finds the company is undervalued (a 34.18% discount), so a **BUY** recommendation is issued.

3. Residual Income Valuation

The core input of the Residual Income Valuation Model is earnings. Residual income is derived by subtracting required income (the threshold return) from earnings, which is used to value the company's profitability.

First, I assume all excess cash will be distributed, and all surplus FCF to Equity will be paid out to common shareholders. The company's free cash flow will only be sufficient to cover dividends in 2028, with no excess cash distribution in other years.

Excess Cash Calculation						
		2025E	2026E	2027E	2028E	2029E
FCF (firm)		129,134	114,578	110,484	115,225	128,092
Interest expense		(124,881)	(124,881)	(124,881)	(14,530)	(131,330)
Effective tax rate		29.08%	29.08%	29.08%	29.08%	29.08%
FCF (equity)		40,563	26,007	21,913	104,920	34,947
Dividend paid out		(53,178)	(53,178)	(53,178)	(53,178)	(53,178)
Excess cash		-12,615	-27,171	-31,265	51,742	-18,231

The Residual Income Valuation Model adopts the same assumptions as the above DCF valuation model. The resulting projected share price is \$48.5, and the current market price is 141% below this projected value, leading to a **BUY** recommendation.

Residual Income Valuation Model						
	2024A	2025E	2026E	2027E	2028E	2029E
Units	Thousands	Thousands	Thousands	Thousands	Thousands	Thousands
Net income to common		85,314	99,119	114,173	207,717	141,113
Common dividend		(53,178)	(53,178)	(53,178)	(53,178)	(53,178)
Common stock repurchases					51,742	
Book value of equity (common + minority)	620,752	652,888	698,829	759,824	966,105	1,054,039
Required income		58,153	61,163	65,467	71,181	90,506
Residual income		27,161	37,955	48,706	136,536	50,607
Present value factor		0.94	0.87	0.82	0.76	0.72
Present value residual income		25,401	33,196	39,839	104,441	36,203
Present value near-term residual income	239,079					
Present value terminal value	1,326,807					
Book value of equity (common + minority)	620,752					
Less: Minority interest	(143)					
Add: Excess cash	(37,563)					
Value of common equity	2,148,932					
Shares outstanding	44,315					
Forecasted per share price	\$ 48.5					
Current stock price	\$ 20.1					
Over/under valuation \$	-28.40					
Over/under valuation %	-141%					
RECOMMENDATION	BUY					

4. Relative Valuation

I selected P/E ratio, TEV/Rev, and TEV/EBITDA as the core industry valuation metrics. Comparable companies were chosen from Capital IQ's recommended peers with similar business operations and complete data, excluding those with insufficient data. Even after excluding extreme value (not presented in table), the company still remains seriously undervalued. All market capitalization figures and metrics are current data sourced from Yahoo Finance.

The relative valuation results consistently indicate that the company's market capitalization is significantly

undervalued, supporting a **BUY** recommendation. The average and median results are nearly identical, and valuations based on TEV metrics reflect a more pronounced undervaluation of Deluxe's market capitalization compared to those using the P/E ratio.

Comparables Valuation Model (Current)						
Company	Market capitalization	Trailing P/E	Forward P/E	TEV	TEV/Rev (TTM)	TEV/EBITDA (TTM)
units	Millions			Millions		
Quad/Graphics, Inc. (NYSE: QUAD)	292	13.67	4.95	835	0.33	5.18
WEX Inc. (NYSE: WEX)	5,020	19.07	8.31	5,110	1.95	5.31
ACI Worldwide, Inc. (NasdaqGS: ACIW)	4,760	18.69	13.23	5,460	3.15	11.26
Corpay, Inc. (NYSE: CPAY)	20,730	20.13	12.02	26,840	6.22	11.75
Clearwater Paper Corporation (NYSE: CLW)	292	11.32	158.73	605	0.39	15.12
Euronet Worldwide, Inc. (NasdaqGS: EEF)	3,140	10.90	6.71	3,570	0.85	5.43
Priority Technology Holdings, Inc. (NasdaqCM: PRTH)	479	11.04	6.43	1,430	1.53	7.36
Average	4,959	14.97	30.05	6,264	2.06	8.77
Median	3,140	13.67	8.31	3,570	1.53	7.36
Deluxe Corporation (NYSE: DLX)	904	11.04	5.69	2,380	1.12	6.3
Implied Value Driver		82	159		2,120	383
Relative \$ Valuation (using Average)		1,226	4,776			
Relative \$ Valuation (using Median)		1,120	1,321			
Relative \$ TEV Valuation (using Average)					4,367	3,356
Relative \$ TEV Valuation (using Median)					3,244	2,816
MVD - Cash					1,476	1,476
Relative \$ Equity Valuation (using Average)					2,891	1,880
Relative \$ Equity Valuation (using Median)					1,768	1,340
Relative O/U Valuation (using Average)		-26.27%	-81.07%		-68.73%	-51.92%
Relative O/U Valuation (using Median)		-19.24%	-31.53%		-48.85%	-32.52%
RECOMMENDATION		BUY	BUY		BUY	BUY

5. Discussion of Similarities/Differences between Approaches

➤ *Similarities*

The three valuation methods stated above (DCF, Residual Income Valuation, and Relative Valuation) all target shareholders' equity value. They all assess the worth of the future economic benefits the company will generate under given cost of capital assumptions, and the three methods align on key inputs (terminal growth rates, profit margins, WACC). As a result, all three methods lead to the conclusion that the company is undervalued and merits a **BUY** rating.

➤ *Differences*

- The DCF Valuation focuses on future cash generation, deriving FCF from EBIT, NOPAT, and adjustments for non-cash items, CAPEX, and net working capital changes. It features the most comprehensive calculation process but is highly dependent on cash flow projections and terminal value assumptions.
- The Residual Income Valuation centers on earnings, calculating residual income as earnings minus required threshold returns. It shares core assumptions with the DCF model but is more responsive to dividends and cash flow while being less affected by terminal value and debt, making it suitable for investors prioritizing current earnings performance.

- The Relative Valuation, by contrast, relies on industry benchmark metrics and compares Deluxe to comparable companies. It uses real-time market data and TEV-based metrics reflecting its focus on relative market positioning rather than intrinsic cash flow or earnings potential. It is simple and intuitive to implement but highly sensitive to the selection of comparable companies and current market sentiment.

6. Sensitivity Checks

I used WACC and terminal growth rate as core variables, this analysis assesses how changes in these factors impact the target share price.

- When the terminal growth rate is fixed, a higher WACC leads to a lower share price valuation (e.g., at a 3.00% terminal growth rate, as WACC rises from 6.43% to 7.43%, the share price drops from \$40.0 to \$23.1 in DCF model).
- When WACC is fixed, a higher terminal growth rate results in a higher share price valuation (e.g., at a 6.93% WACC, as the terminal growth rate increases from 2.50% to 3.50%, the share price rises from \$45.0 to \$53.0 in Residual income model).
- The benchmark combination of both valuation models corresponds to a target share price of \$30.5 and \$48.5 respectively, which aligns with the final valuation result.
- The highlighted areas indicate that even slight deviations in the assumed terminal growth rate and WACC fall within a reasonable range, and this will not affect the final conclusion that Deluxe is undervalued.

Sensitivity Analysis - DCF method							
WACC	Terminal growth rate						
	\$	30.5	2.50%	2.75%	3.00%	3.25%	3.50%
		6.43%	31.7	35.6	40.0	45.2	51.2
		6.68%	27.7	31.1	34.9	39.4	44.5
		6.93%	24.2	27.1	30.5	34.3	38.7
		7.18%	21.0	23.6	26.6	29.9	33.7
		7.43%	18.2	20.5	23.1	26.1	29.4

Sensitivity Analysis - Residual income method							
WACC	Terminal growth rate						
	\$	48.5	2.50%	2.75%	3.00%	3.25%	3.50%
		6.43%	48.4	50.5	52.9	55.7	59.0
		6.68%	46.6	48.5	50.6	53.0	55.8
		6.93%	45.0	46.6	48.5	50.6	53.0
		7.18%	43.5	45.0	46.6	48.5	50.6
		7.43%	42.2	43.5	45.0	46.7	48.5

Part 4: Recommendation

Deluxe is a mature, cash-generative established company undergoing a gradual transformation from its traditional check business to a platform-based enterprise centered on payment services and data solutions. Projections for the three financial statements have incorporated the transformation direction outlined by management, the structural shift from product to service revenue, moderate revenue growth in line with U.S. nominal GDP, and modest margin improvements driven by cost optimization and restructuring, while ensuring logical consistency across the income statement, balance sheet, and cash flow statement.

The three valuation methods adopted consistent assumptions regarding growth rates, profit margins, and WACC, yielding highly consistent conclusions from cash flow, earnings, and market pricing perspectives, that is **the current stock price undervalues Deluxe.**

- **DCF Model:** Under the assumptions of a WACC of approximately 6.93% and a terminal growth rate of 3%, the intrinsic value per share is estimated at \$30.52. Relative to the current stock price of \$20.09, this implies a discount of around 34%, supporting a Buy rating.
- **Residual Income Model:** Leveraging the same fundamental assumptions as the DCF model, this approach values the company based on earnings and residual income rather than cash flow, resulting in a higher target price of \$48.5, which is also significantly above the current market price.
- **Relative Valuation Model:** Key multiples were compared with comparable companies featuring similar business structures and complete data. Even after excluding outliers, Deluxe trades at a significant discount in both earnings and enterprise value multiples, with TEV-based metrics reflecting a more pronounced undervaluation.

However, Deluxe carries a relatively **high leverage ratio and faces significant debt maturities in 2029.** The success of its transformation hinges on the execution of payment and data businesses, the sustainability of cost control, and the pace of decline in its traditional check business, rather than mere cost-cutting without appropriate investments. If interest rates remain high for an extended period, the SMB environment deteriorates, or the rollout of digital products falls short of expectations, both earnings and valuation could be adversely impacted.

Under reasonable and relatively conservative assumptions, all three valuation methods indicate moderate to significant undervaluation of Deluxe's stock. We therefore issue a **BUY** recommendation, suitable for **long-term investors willing to accept moderate volatility** and bear the risks associated with the company's transformation and refinancing.